Student: Xirong Zhang **Date:** 11/30/25	**Instructor:** Dmitriy Ivanov **Course:** Math 120 Fall 2025	**Assignment:** Section 1.3

1. Compute $\mathbf{u}+\mathbf{v}$ and $\mathbf{u}-3\mathbf{v}$.

$$\mathbf{u}=\begin{bmatrix}-3\\6\end{bmatrix},\ \mathbf{v}=\begin{bmatrix}-5\\4\end{bmatrix}$$

$\mathbf{u}+\mathbf{v}=\begin{bmatrix}-8\\10\end{bmatrix}$ (Simplify your answer.)

$\mathbf{u}-3\mathbf{v}=\begin{bmatrix}12\\-6\end{bmatrix}$ (Simplify your answer.)

2. Write a system of equations that is equivalent to the given vector equation.

$$x_1\begin{bmatrix}3\\-2\end{bmatrix}+x_2\begin{bmatrix}7\\5\end{bmatrix}+x_3\begin{bmatrix}-5\\1\end{bmatrix}=\begin{bmatrix}0\\0\end{bmatrix}$$

Choose the correct answer below.

- **A.** $\begin{aligned}3x_1+7x_2+5x_3&=0\\2x_1+5x_2+x_3&=0\end{aligned}$
- **B.** $\begin{aligned}3x_1+7x_2+5x_3&=0\\2x_1+5x_2&=0\end{aligned}$
- **C.** $\begin{aligned}3x_1+7x_2-5x_3&=0\\-2x_1+5x_2+x_3&=0\end{aligned}$
- **D.** $\begin{aligned}3x_1+7x_2-5x_3&=0\\-2x_1+5x_2&=0\end{aligned}$

3. Write a vector equation that is equivalent to the given system of equations.

$$\begin{aligned}x_2+3x_3&=0\\5x_1+9x_2-x_3&=0\\-x_1+5x_2-9x_3&=0\end{aligned}$$

Choose the correct answer below.

- **A.** $x_1\begin{bmatrix}0\\5\\-1\end{bmatrix}+x_2\begin{bmatrix}3\\-1\\-9\end{bmatrix}+x_3\begin{bmatrix}1\\9\\5\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}$
- **B.** $x_1\begin{bmatrix}0\\5\\-1\end{bmatrix}+x_2\begin{bmatrix}1\\9\\5\end{bmatrix}+x_3\begin{bmatrix}3\\-1\\-9\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}$
- **C.** $x_1\begin{bmatrix}0\\5\\-1\end{bmatrix}+x_2\begin{bmatrix}1\\9\\5\end{bmatrix}+x_3\begin{bmatrix}0\\0\\0\end{bmatrix}=\begin{bmatrix}3\\-1\\-9\end{bmatrix}$
- **D.** $x_1\begin{bmatrix}1\\9\\5\end{bmatrix}+x_2\begin{bmatrix}0\\5\\-1\end{bmatrix}+x_3\begin{bmatrix}3\\-1\\-9\end{bmatrix}=\begin{bmatrix}0\\0\\0\end{bmatrix}$

4. Determine if **b** is a linear combination of the vectors formed from the columns of the matrix A.

$$A = \begin{bmatrix} 1 & -5 & 3 \\ 0 & 4 & 7 \\ -2 & 10 & -6 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 2 \\ -5 \\ -2 \end{bmatrix}$$

Choose the correct answer below.

- **A.** Vector **b** is a linear combination of the vectors formed from the columns of the matrix A. The pivots in the corresponding
- **B.** Vector **b** is a linear combination of the vectors formed from the columns of the matrix A. The pivots in the corresponding echelon matrix are in the first entry in the first column and the third entry in the second column, and the third entry in the third column.
- **C.** Vector **b** is not a linear combination of the vectors formed from the columns of the matrix A.
- **D.** Vector **b** is a linear combination of the vectors formed from the columns of the matrix A. The pivots in the corresponding echelon matrix are in the first entry in the first column, the second entry in the second column, and the third entry in the fourth column.

5. Determine if **b** is a linear combination of the vectors formed from the columns of the matrix A.

$$A = \begin{bmatrix} 1 & -6 & -4 \\ 0 & 2 & 5 \\ 2 & -12 & 7 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 13 \\ -7 \\ 8 \end{bmatrix}$$

Choose the correct answer below.

- **A.** Vector **b** is a linear combination of the vectors formed from the columns of the matrix A. The pivots in the corresponding echelon matrix are in the first entry in the first column, the second entry in the second column, and the third entry in the third column.
- **B.** Vector **b** is a linear combination of the vectors formed from the columns of the matrix A. The pivots in the corresponding echelon matrix are in the first entry in the first column and the second entry in the second column. There are no pivots in the third and fourth columns.
- **C.** Vector **b** is not a linear combination of the vectors formed from the columns of the matrix A. The pivots in the corresponding echelon matrix are only in the first entry in the first column and the second entry in the second column. There are no pivots in the third and fourth columns.
- **D.** Vector **b** is not a linear combination of the vectors formed from the columns of the matrix A. There is no solution to the linear system corresponding to the matrix $\begin{bmatrix} A & \mathbf{b} \end{bmatrix}$.

6. List five vectors in Span$\{\mathbf{v}_1, \mathbf{v}_2\}$. Do not make a sketch.

$$\mathbf{v}_1 = \begin{bmatrix} 8 \\ 1 \\ -5 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} -6 \\ 3 \\ 0 \end{bmatrix}$$

List five vectors in Span$\{\mathbf{v}_1, \mathbf{v}_2\}$.

$$\begin{bmatrix} 8 \\ 1 \\ -5 \end{bmatrix}, \begin{bmatrix} 16 \\ 2 \\ -10 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} -8 \\ -1 \\ 5 \end{bmatrix}, \begin{bmatrix} -16 \\ -2 \\ 10 \end{bmatrix}$$

(Use the matrix template in the math palette. Use a comma to separate vectors as needed. Type an integer or a simplified fraction for each vector element. Type each answer only once.)

7. Let $\mathbf{u} = \begin{bmatrix} 6 \\ -1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 6 \\ 1 \end{bmatrix}$. Show that $\begin{bmatrix} x \\ c \end{bmatrix}$ is in Span {**u**, **v**} for all x and c.

How can it be shown that a vector **b** is in Span {**u**, **v**}?

- **A.** Determine if the system containing **u**, **v**, and **b** is consistent. If the system is consistent, **b** might be in Span {**u**, **v**}.
- **B.** Determine if the system containing **u**, **v**, and **b** is consistent. If the system is inconsistent, then **b** is in Span {**u**, **v**}.
- **C.** Determine if the system containing **u**, **v**, and **b** is consistent. If the system is consistent, then **b** is in Span {**u**, **v**}.
- **D.** Determine if the system containing **u**, **v**, and **b** is consistent. If the system is consistent, then **b** is not in Span {**u**, **v**}.

Let $\mathbf{b} = \begin{bmatrix} x \\ c \end{bmatrix}$. Find the augmented matrix $\begin{bmatrix} \mathbf{u} & \mathbf{v} & \mathbf{b} \end{bmatrix}$.

$$\begin{bmatrix} 6 & 6 & x \\ -1 & 1 & c \end{bmatrix}$$

How is a system determined as consistent?

- **A.** A system is consistent only if all of the variables equal each other.
- **B.** A system is consistent if there are no solutions.
- **C.** Solve for the variables after setting the equations equal to 0.
- **D.** A system is consistent if there is one solution or infinitely many solutions.

Row reduce the augmented matrix to its reduced echelon form.

$$\begin{bmatrix} 6 & 6 & x \\ -1 & 1 & c \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & \frac{(x-6c)}{12} \\ 0 & 1 & \frac{(x+6c)}{12} \end{bmatrix}$$

The system is consistent for all values of x and c. So it can be stated that $\begin{bmatrix} x \\ c \end{bmatrix}$ is in Span {**u**, **v**} for all values of x and c.

8. Construct a 3×3 matrix A, with nonzero entries, and a vector **b** in $\mathbb{R}^3$ such that **b** is not in the set spanned by the columns of A.

Choose the correct answer below.

- **A.** $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 4 & 5 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 6 \end{bmatrix}$
- **B.** $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix}$
- **C.** $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 1 \\ 3 & 3 & 2 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$
- **D.** $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 3 & 3 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$

9. Determine whether the statement below is true or false. Justify the answer.

The points in the plane corresponding to $\begin{bmatrix} -2 \\ 5 \end{bmatrix}$ and $\begin{bmatrix} -5 \\ 2 \end{bmatrix}$ lie on a line through the origin.

Choose the correct answer below.

- **A.** The statement is false. $\begin{bmatrix} -2 \\ 5 \end{bmatrix}$ and $\begin{bmatrix} -5 \\ 2 \end{bmatrix}$ correspond to arrows that meet at right angles.
- **B.** The statement is true. $\begin{bmatrix} -2 \\ 5 \end{bmatrix}$ and $\begin{bmatrix} -5 \\ 2 \end{bmatrix}$ are on a line through the origin since $\begin{bmatrix} -5 \\ 2 \end{bmatrix}$ is a positive multiple of $\begin{bmatrix} -2 \\ 5 \end{bmatrix}$.
- ✔ **C.** The statement is false. If $\begin{bmatrix} -2 \\ 5 \end{bmatrix}$ and $\begin{bmatrix} -5 \\ 2 \end{bmatrix}$ were on the line through the origin, then $\begin{bmatrix} -5 \\ 2 \end{bmatrix}$ would be a multiple of $\begin{bmatrix} -2 \\ 5 \end{bmatrix}$, which is not the case.
- **D.** The statement is true. $\begin{bmatrix} -2 \\ 5 \end{bmatrix}$ and $\begin{bmatrix} -5 \\ 2 \end{bmatrix}$ correspond to arrows pointing in precisely opposite directions and so are on the same line through the origin.

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10. Determine whether the statement below is true or false. Justify the answer.

An example of a linear combination of vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ is the vector $\frac{1}{2}\mathbf{v}_1$.

Choose the correct answer below.

- **A.** The statement is true because $\frac{1}{2}\mathbf{v}_1 = \frac{1}{4}\mathbf{v}_1 + \frac{1}{4}\mathbf{v}_2$.
- **B.** The statement is false because a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$ needs to involve both of those vectors.
- **C.** The statement is false because $\frac{1}{2}\mathbf{v}_1$ cannot be expressed as a multiple of $\mathbf{v}_2$.
- ✔ **D.** The statement is true because $\frac{1}{2}\mathbf{v}_1 = \frac{1}{2}\mathbf{v}_1 + 0\mathbf{v}_2$.